CS432: Databases

Database Design and the E-R Model

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13

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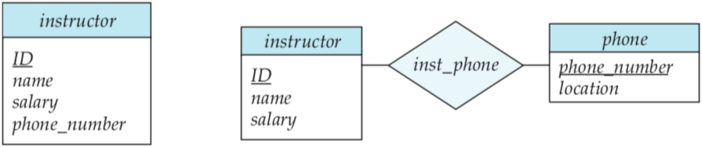
Recap

- Entity-Relationship Diagrams
- Reduction to Relational Schemas
- Entity-Relationship Design Issues

Entity-Relationship Design Issues

Use of Entity Sets versus Attributes



Attribute *name* (of an instructor) as an entity?

Two common mistakes

- To use the primary key of an entity set as an attribute of another entity set.
- For example,

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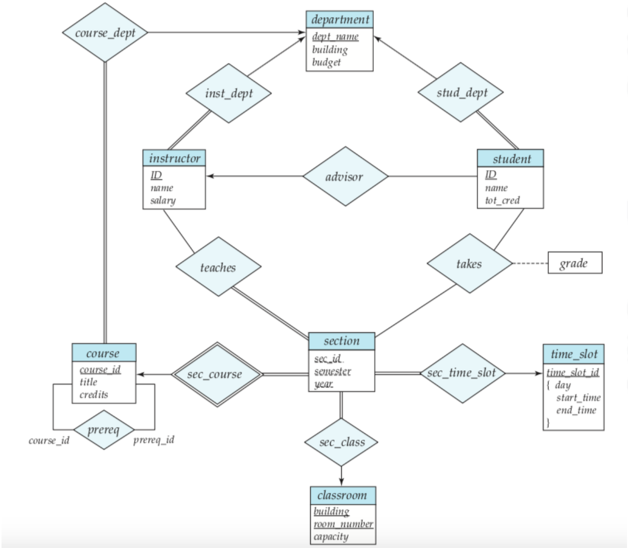
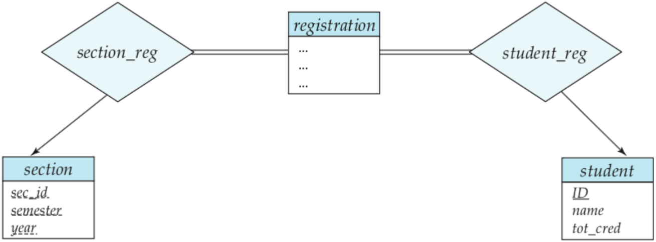
- For example, ID of a *student* as an attribute of an *instructor*, even if the instructor advises only one student.
- To designate the primary-key attributes of the related entity sets as attributes of the relationship set.
- For example, ID of *student* and ID of *instructor* should not appear as attributes of the relationship advisor.

Entity-Relationship Design Issues



Entity-Relationship Design Issues

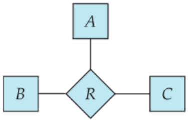
Use of Entity Sets versus Relationship Sets

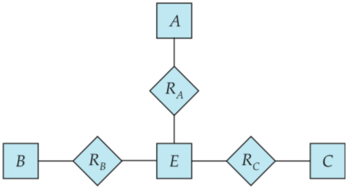


Entity-Relationship Design Issues

Binary versus n-ary Relationship Sets

Mother and father entity sets relating a child entity.





For each relationship (a_i , b_i , c_i) in the relationship set R, we create a new entity e_i in the entity set E.

- (e_i,a_i) in R_A.
- (e_i,b_i) in R_B.
- (e_i,c_i) in R_C.

Entity-Relationship Design Issues

Binary versus n-ary Relationship Sets

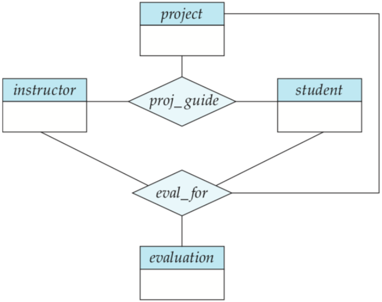
Conversion to Binary relationships may lead to:

- Increase in the complexity of the design and overall storage requirements.
- An n-ary relationship set shows more clearly that several entities participate in a single relationship.
- Not possible to translate constraints on the ternary relationship into constraints on the binary relationships.

Entity-Relationship Design Issues

Binary versus n-ary Relationship Sets

Can we split *proj_guide* into binary relationships between instructor and project and between instructor and student?



Entity-Relationship Design Issues

Placement of Relationship Attributes

The cardinality ratio of a relationship can affect the placement of relationship attributes.

- Attributes of a one-to-many relationship set can be repositioned to only the entity set on the "many" side of the relationship.
- For one-to-one relationship sets the relationship attribute can be associated with either one of the participating entities.

Dependent on the characteristics of the enterprise being modeled.

Extended E-R Features

Specialization

The process of designating subgroupings within an entity set is called specialization.

- Example:

Extended E-R Features

Specialization

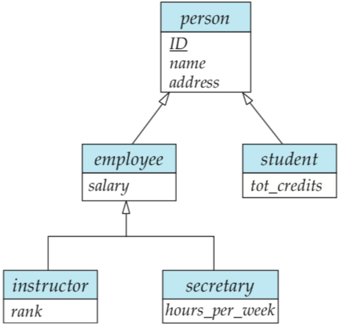
The process of designating subgroupings within an entity set is called specialization.

- Example: **Graduate** and **undergraduate** entity sets in **student** entity set.
- Each type includes all the attributes of entity set **student** plus possibly additional attributes.
- Student: ID, name, address, and tot cred
  Graduate: Student + office_number
  Undergraduate = Student + residential_college

Specialization is shown by a hollow arrow-head pointing from the specialized entity to the other entity.

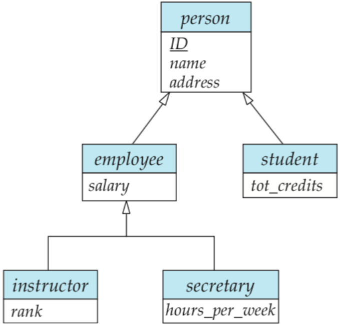
Extended E-R Features

Specialization



Extended E-R Features

Specialization



Overlapping specialization
Disjoint specialization

Extended E-R Features

Generalization

The database designer may have first identified:

- **Instructor** entity set with attributes instructor id, instructor name, instructor_salary, and rank.
- **Secretary** entity set with attributes secretary_id, secretary_name, secretary_salary and hour_per_week.

The commonality in the above attributes can be expressed by generalization.

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Higher-level entity set and one or more lower-level entity sets

Extended E-R Features

Attribute Inheritance

- The attributes of the higher-level entity sets are said to be inherited by the lower-level entity sets.
- A lower-level entity set (or subclass) also inherits participation in the relationship sets in which its higher-level entity (or superclass) participates.

Extended E-R Features

Constraints on Generalizations

Constraint involves determining which entities can be members of a given lower-level entity set.

- **Condition-defined**: student_type attribute in student entity.
- **User-defined**: The database user assigns entities to a given entity set
- **Disjoint**: An entity belong to no more than one lower-level entity set
- **Overlapping**: An entity can belong to more than one lower-level entity set.
- **Completeness**: Whether or not an entity in the higher-level entity set must belong to at least one of the lower-level entity sets.
  - Total generalization or specialization
  - Partial generalization or specialization

Extended E-R Features

Aggregation

Each instructor guiding a student on a project is required to file a monthly evaluation report.

Extended E-R Features

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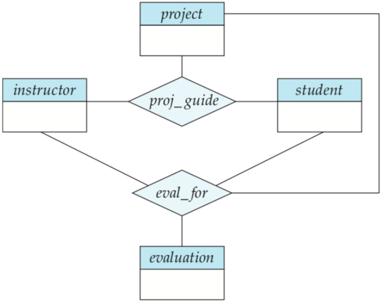

Extended E-R Features

Aggregation

Aggregation is an abstraction through which relationships are treated as higher-level entities.

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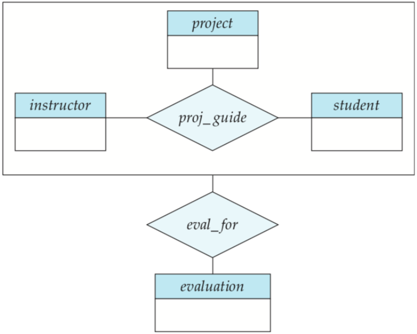

Reduction to Relation Schemas

Representation of Generalization

One possible solution:

- Create a schema for the higher-level entity set.
- For each lower-level entity set create a schema that includes:
  - an attribute for each of the attributes of that entity set
  - one for each attribute of the primary key of the higher-level entity set

person (<u>ID</u>, name, street, city)
employee (<u>ID</u>, salary)
student (<u>ID</u>, tot_cred)

Representation of Generalization

Another possible solution:

- Do not create a schema for the higher-level entity set.
- For each lower-level entity set create a schema that includes:
  - an attribute for each of the attributes of that entity set
  - one for each attribute of the higher-level entity set

employee (<u>ID</u>, name, street, city, salary)
student (<u>ID</u>, name, street, city, tot_cred)

Potential problems with the second approach?

Reduction to Relation Schemas

Representation of Aggregation

The schema for the relationship set **eval_for** includes an attribute for each attribute in the primary keys of the entity set **evaluation**, and for the relationship set **proj_guide**.

Other Aspects of Database Design

- Data Constraints and Relational Database Design
- Usage Requirements: Queries, Performance
- Authorization Requirements
- Data Flow, Workflow
- Other Issues in Database Design

Acknowledgments/Contributions

- Some of the images utilized in these slides are subject to copyright - Abraham Silberschatz, Henry Korth, and S. Sudarshan. Database System Concepts. 6th Edition, McGraw-Hill Education
- Contributor 2024-25, 2025-26 - Yogesh K. Meena, IIT Gandhinagar